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SUMMARY

AI Engineer with a strong background in Applied Mathematics, specializing in the design and deployment of end-to-end LLM-powered systems. Proven experience architecting and deploying end-to-end AI solutions for complex challenges in large-scale data analysis, including classification and semantic search.

WORK EXPERIENCE

Data Scientist

Jan. 2025 – June 2025

Invictus Growth Partners

San Mateo, CA

- Engineered and deployed an LLM-powered framework for the classification of large-scale, unstructured data.
- Built a scalable system for semantic search and analysis, developing custom re-ranking algorithms to enhance result relevance.
- Architected end-to-end AI systems, taking full ownership of the project lifecycle from initial concept to production-ready deployment.

AI Training Specialist

Jul. 2024 – Jan. 2025

DataAnnotation & Outlier AI

Freelance

- Enhanced AI models by correcting coding and mathematical responses, improving accuracy and reliability.
- Identified and rectified hallucinations in AI outputs, contributing to significant error rate reduction.

Graduate Teaching Assistant

Aug. 2021 – May 2024

University of Colorado Boulder

Boulder, CO

PROJECTS

MathBuddy: AI-Powered Math Tutor | *Next.js, FastAPI, Python*

Aug. 2024 – Sept. 2024

- Engineered a full-stack AI tutor, leveraging GPT-4o for interactions and GPT-3.5-Turbo for result extraction and difficulty estimation.
- Implemented serverless architecture with Next.js frontend and FastAPI backend, integrating OpenAI and Wolfram Alpha APIs to enhance problem-solving capabilities.

Tic-Tac-Toe with Alpha-Beta Pruning | *Python, Pygame, NumPy*

June 2024

- Developed a Tic-Tac-Toe game featuring an AI opponent using the Alpha-Beta Pruning Minimax algorithm.
- Improved AI decision-making speed by 40% by reducing evaluated nodes, enhancing gameplay experience.

ODE Solution via PINNs | *Python, TensorFlow, SciPy*

Oct. 2023 – Dec. 2023

- Engineered and trained a Physics-Informed Neural Network (PINN) to accurately model the damped pendulum problem, demonstrating the successful application of a cutting-edge deep learning technique to solve complex ODEs.

EDUCATION

University of Colorado Boulder

Aug. 2021 – May 2024

*M.S. Applied Mathematics, **Focus:** Data Science & Machine Learning*

Boulder, CO

Lake Forest College

Aug. 2016 – May 2020

B.A. Mathematics, B.A. Economics

Lake Forest, IL

Honors: Summa Cum Laude, Phi Beta Kappa

TECHNICAL SKILLS

Languages & Databases: Python (Expert), R, SQL, C++, JavaScript

AI/ML: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, SciPy, NLP, PINNs

Tools & Web Dev: FastAPI, Next.js, Git, Docker, Power BI, $\LaTeX$